

PRATISHTHA SONI

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SUMMARY

- Software Engineer with a Master's in Computer Science and over 5 years of experience applying **AI and machine learning** to solve complex business problems and enhance **customer experiences**.
- Proven track record in **Generative AI (GenAI)**, **Natural Language Processing (NLP)**, and **LLM-based automation** to drive business insights and operational efficiency.
- Published researcher in **Machine Learning**, with papers presented at prestigious conferences such as **AIMHC-IEEE** and **EC BIOS-IEEE** in 2024, focusing on innovative applications in the Healthcare and social services domain.
- Contributed to agricultural technology advancements, with publication on ***Agricultural Crop Yield Prediction Using Artificial Intelligence and Satellite Imagery*** in **Eurasian Journal of Analytical Chemistry** 2019.
- Recognized for excellence, awarded Spot Award for the ***Automation of Data Extraction*** in L&T Technology Services Limited in 2021.

TECHNICAL SKILLS

Languages & Frameworks : Python, C++, Fask, API Development, Docker, Kubernetes, CSS, HTML

Cloud Services : AWS (S3,Lambda,DynamoDb, Opensearch, API, VPC, EC2, EKS, Bedrock), Azure (AI Studio, AzureAI), HuggingFace

ML/AI Technologies: Prompt Engineering, LLMs, RAG, Transfer Learning, Deep Learning, NLP, Computer Vision, TensorFlow, PyTorch

Data Engineering: Hadoop, Spark, PySpark, HDFS, SQL, Snowflake

PROFESSIONAL EXPERIENCE

Software Engineer (AI/Backend), StarTek, Denver

September 2024 - Present

- Automated QA workflows using Prompt Engineering and LLMs, to analyze customer-agent transcriptions, enhancing response accuracy by 90% and incorporating KPI-driven evaluation metrics to assess agent performance and customer satisfaction.
- Designed and implemented a production AI system with RAG integration using AWS S3, Bedrock, Lambda, OpenSearch, API Gateway, and OAuth, delivering customized feedback to agents based on client compliance knowledge articles and datasets.
- Automated CI/CD pipelines (CodeCommit, CodeBuild, CodeDeploy) with Docker containerization, for a serverless call analytics system, providing real-time insights for 5,000 concurrent users, boosting agent query efficiency by 25%, and reducing costs by 40%.

Research Associate, University of Southern California (USC), California

February 2025 - Present

- Leveraging Agentic AI in Dr. Nick Vyas's (*Founder & Executive Director*) research paper to eliminate manual intervention, enhance autonomous decision-making, and drive efficiency in supply chain optimization.

ML Research Associate, California State University Fullerton, CA

March 2023 - May 2024

- Developed scalable predictive and statistical classification models using SSD MobileNet V2 and transfer learning to analyze real-time food images and provide guided cooking recommendations.
- Utilized TensorFlow, Keras, PyTorch, and OpenCV frameworks for data modeling and data preprocessing.

Senior Python Developer, ValueLabs, India

February 2022 - Aug 2022

- Developed and tested REST APIs for website integration, reducing API response time by 40%, leading to better user experience.
- Optimized the deployment of Kafka producer and consumer scripts to enhance customer notifications, automating live data integration across multiple topics and efficiently resolving issues for over 1,000 requests.

Software Engineer, L&T Technology Services, India

June 2019 - February 2022

- Identified patterns from unstructured documents and extracted features by leveraging Natural Language Processing (NLP), led to 70% accuracy and improved the quality and reliability of the data, saving 25 hours of manual work per week.
- Built an internal tool using Power BI and SQL to track document processing time, total count, and efficiency KPIs, enabling data-driven decision-making and improving business strategy effectiveness.
- Automated web scraping of selected features from oil and engineering websites and applied statistical analysis with 80% accuracy using Selenium and following Agile principles for iterative improvements.
- Executed tabular data extraction and export to a predefined report format by executing data integration scripts using SSIS to read .json and .xlsx files, enhancing scalability and reducing manual data entry processes by 40%.
- Developed relationships across multiple Excel files using Graph Knowledge and OWL, leveraging GraphQL for structured data querying and visualizing connections

ACADEMIC PROJECTS

NYC Automated Traffic Volume Counts (*pandas, hadoop, java, spark, pyspark, SQL, HDFS*)

- Utilized big data tools to analyze diverse data trends through Spark queries on GCP within the Hadoop framework.

Crop Yield Prediction using Artificial Intelligence and Satellite Imagery (*opencv, scikit-image, tensorflow, pytorch*)

- Implemented a rule-based system to predict crop yield from a pool of GIS data by applying ANN and CNN with 89% accuracy.

EDUCATION

California State University Fullerton, USA - *Master of Science, Computer Science*

August 2022 - May 2024

SRM Institute of Science and Technology, India - *Bachelor of Technology, Computer Science and Engineering*

July 2015 - May 2019